Two uses: change exposure or reduce borrowing cost

LECTURE 11 · REVIEW NOTES

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There are two decisions I want you to keep separate. With debt already outstanding, the CFO may want to change interest-rate exposure. With new borrowing, the CFO may compare a direct loan with a borrowing-plus-swap package to reduce cost. We combine the debt and swap cash flows in both cases, but we judge the result against a different objective.

Start with an existing floating liability

Our company owes USD 2,000 at SOFR + 1%. It wants a predictable interest bill, so it enters a swap paying 4% fixed and receiving SOFR on the same notional and semiannual dates. The debt payment goes to the lender; the swap is a separate contract.

$$\text{Net swap CF} = \text{Swap received} - \text{Swap paid}$$

$$\text{Final cost} = \text{Debt paid} - \text{Net swap CF}$$

At SOFR = 3%, debt interest is USD 40. The swap pays USD 40 fixed and receives USD 30 floating, so its net cash flow is -USD 10. Final borrowing cost is 40 - (-10) = USD 50. Paying on the swap has not defeated the hedge: the debt payment was lower too.

Payment	SOFR used	Debt paid	Swap net CF	Final cost
t=0.5	3%	$40	-$10	$50
t=1.0	5%	$60	+$10	$50
t=1.5	6%	$70	+$20	$50
t=2.0	2%	$30	-$20	$50

$$(\text{SOFR} + 1\%) + 4\% - \text{SOFR} = 5\%, \qquad 2000 \times \frac{5\%}{2} = 50$$

What the CFO has purchased

The company gives up the benefit of falling rates to obtain a 5% fixed annual cost. The debt's SOFR payment cancels the swap's SOFR receipt. This cancellation requires matching notionals, timing and benchmarks. We use annual rates and semiannual payments.

New debt: compare the preferred direct market

Now neither firm has committed to the new financing. A wants fixed-rate debt; B wants floating-rate debt. Before deciding what to borrow, compare the rates available to each firm in both markets.

Firm	Direct fixed rate	Direct floating rate	Wants
A	8%	SOFR + 1%	Fixed
B	9%	SOFR + 3%	Floating

A borrows at SOFR + 1%; B borrows at 9% fixed. They then arrange a two-year swap on USD 2,000 each: A pays B 7.5% fixed and B pays A SOFR + 1%, with matched semiannual dates. Neither exchanges principal under the swap.

Add what each firm pays; subtract what it receives

$$A:\ (\text{SOFR}+1\%)+7.5\%-(\text{SOFR}+1\%)=7.5\%$$

$$B:\ 9\%+(\text{SOFR}+1\%)-7.5\%=\text{SOFR}+2.5\%$$

A now has the fixed exposure it wanted at 7.5%, below its direct fixed rate of 8%. B has the floating exposure it wanted at SOFR + 2.5%, below its direct floating rate of SOFR + 3%. Each saves 0.5 percentage points annually, or USD 10 per year on USD 2,000.

Why is there room for both to save?

The difference between the firms' fixed borrowing quotes is one percentage point; the difference between their floating quotes is two. Under these assumed terms, routing the borrowing through the other market and sharing the difference via the swap produces one percentage point of combined annual savings.

The correct comparison

Compare each final rate with the direct rate in the market the firm actually wanted. These are illustrative borrowing and swap terms, not a promise of costless savings. Fees, collateral requirements and counterparty risk can change the economics.

A realized SOFR path makes the exposure visible

Use the same A/B financing package. SOFR observed at t=0, 0.5, 1.0, 1.5 and 2.0 is 3%, 4%, 2%, 5% and 4%. Each payment uses the rate from the preceding observation. The final 4% is unused because there is no remaining semester.

Firm A: the floating debt and floating receipt cancel

Let F denote the semester's floating amount: USD 2,000 x (SOFR + 1%) / 2. A pays F on the debt, pays USD 75 on the fixed swap leg and receives F on the floating swap leg. Its final cost is always USD 75.

Payment	Debt paid	Swap paid	Swap received	Final cost
t=0.5	$40	$75	$40	$75
t=1.0	$50	$75	$50	$75
t=1.5	$30	$75	$30	$75
t=2.0	$60	$75	$60	$75

Firm B: the final cost remains floating

B pays USD 90 on its fixed debt, pays F through the swap and receives USD 75 fixed. Final cost is 90 + F - 75, which varies with SOFR.

Payment	Debt paid	Swap paid	Swap received	Final cost
t=0.5	$90	$40	$75	$55
t=1.0	$90	$50	$75	$65
t=1.5	$90	$30	$75	$45
t=2.0	$90	$60	$75	$75

A pays USD 300 total interest rather than USD 320 at its direct 8% fixed rate. B pays USD 240 rather than USD 260 at SOFR + 3% on this path. Each saves USD 20 over two years. B's total interest still depends on SOFR; its spread saving does not.

Check the two swap sides

A's swap net cash flow is F - 75; B's is 75 - F. They add to zero for each payment date. This check concerns the swap only, not the payments to the original lenders.

A bank can intermediate the two exposures

Consider a separate package with USD 2,000 notionals and semiannual dates. Atlas chooses 7% fixed borrowing, pays SOFR + 0.5% to the bank and receives 6% fixed. Beacon borrows at SOFR + 1%, pays 6.5% fixed to the bank and receives SOFR - 0.5%.

$$\text{Atlas cost} = 7\% + (\text{SOFR} + 0.5\%) - 6\% = \text{SOFR} + 1.5\%$$

$$\text{Beacon cost} = (\text{SOFR} + 1\%) + 6.5\% - (\text{SOFR} - 0.5\%) = 8\%$$

Atlas wants floating financing and compares this with a direct SOFR + 2% quote. Beacon wants fixed financing and compares with 8.5%. Each saves 0.5 percentage points under the given terms.

Express the swap net in rates: received minus paid

Party	Net swap rate	At SOFR = 4%
Atlas	6% - (SOFR + 0.5%)	+1.5%
Beacon	(SOFR - 0.5%) - 6.5%	-3%
Bank	[(SOFR + 0.5%) + 6.5%] - [6% + (SOFR - 0.5%)]	+1.5%

The bank receives the payments made by both clients and pays the amounts each client receives. The SOFR terms cancel, leaving 1.5% annually on the matched notional, before expenses and credit losses. At 4% SOFR the three net swap rates sum to zero: +1.5% - 3% + 1.5% = 0.

To convert a net annual rate into a semester cash flow, multiply it by USD 2,000 and divide by two. The bank receives USD 15 net. Atlas's final annual cost is 5.5%, while Beacon's is 8%. Do not confuse those borrowing costs with the net swap rates.

What to retain

An existing liability can be transformed without rewriting the debt. New borrowing can be combined with a swap to seek a lower rate in the desired exposure. For either use, write debt plus swap paid minus swap received; cancel matching SOFR terms and compare the result with the right alternative.